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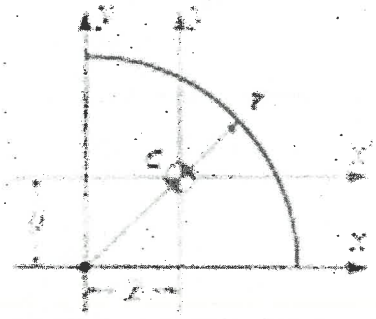
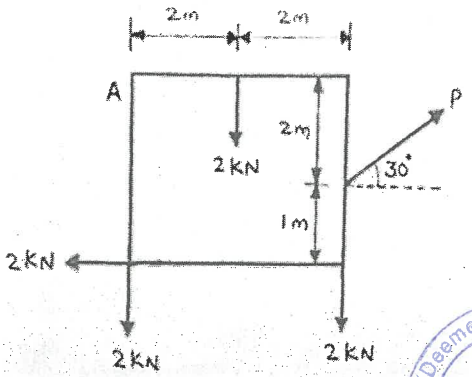
THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ENGINEERING MECHANICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Draw a free body diagram for a beam supported at both ends with a downward force of 200 N applied at its center. Identify all forces acting on the beam.	L3	CO1	**02**
	b.	A box weighing 80 N is placed on a surface with a coefficient of static friction of 0.5. Calculate the maximum static frictional force acting on the box.	L3	CO2	**02**
	c.	Find the centroid C = $(\bar{x}, \bar{y})$ for the given quarter circle of radius r. 	L3	CO3	**02**
	d.	A projectile is launched with an initial velocity of 20 m/s at an angle of 30° to the horizontal. Solve for the maximum height reached by the projectile.	L3	CO5	**02**
	e.	If a 12 kg mass is dropped from a height of 10 meters, calculate the work done by the gravitational force during its fall.	L3	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Solve for the value of force P, so that moment about point A is zero as shown in the figure 	L3	CO1	**05**
		OR			

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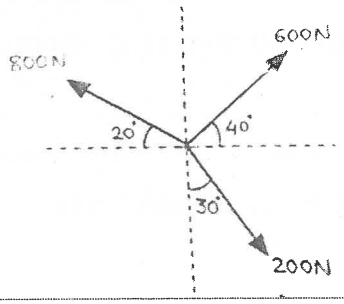
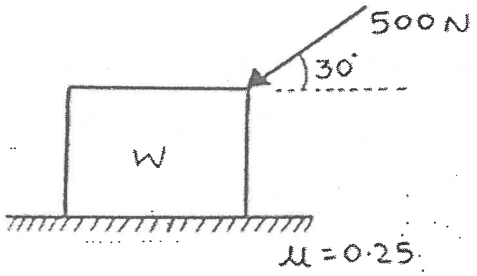
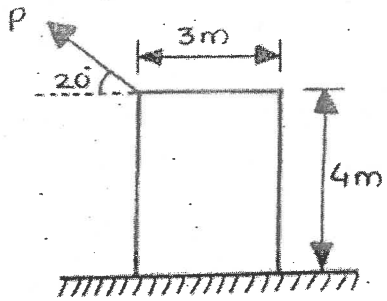
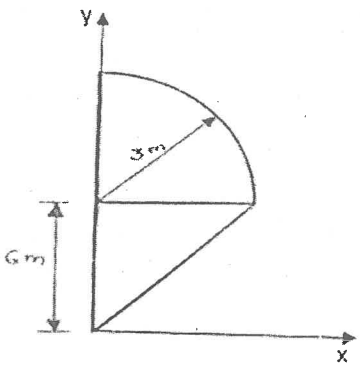
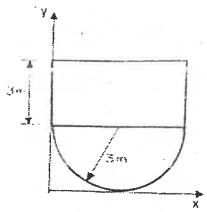
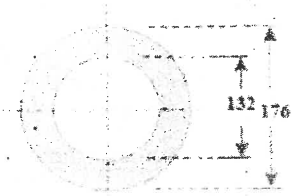
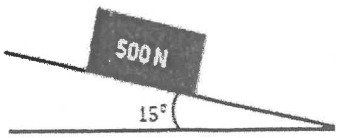
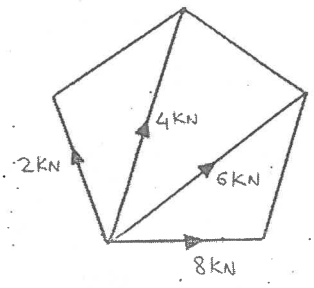
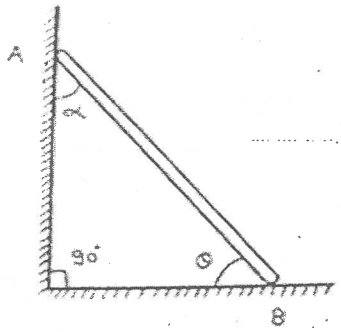
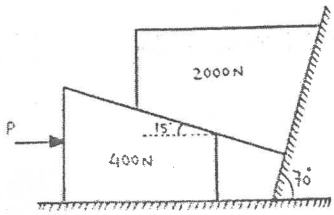
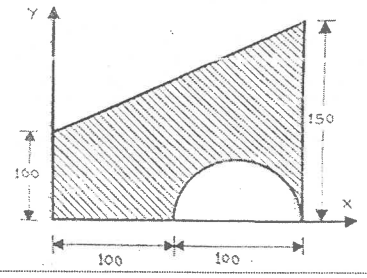
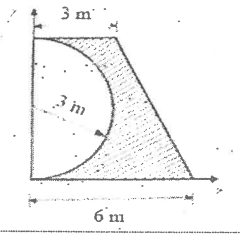
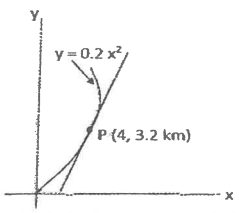
	b.	Solve for the resultant of the force system shown in figure. 	L3	CO1	**05**
3.	**a.**	Find the value of weight "W", if the body is in limiting equilibrium condition. 	L3	CO2	**05**
		OR			
	b.	A block of 700 N is resting on a floor as shown in figure. Determine the value of force P which will cause the block to move. Take $\mu = 0.3$ 	L3	CO2	**05**
4.	**a.**	Calculate the centroid for the given figure with respect to the reference x and y axis. 	L3	CO3	**05**
		OR			

	b.	Find the centroid for the given shape with respect to reference X and Y axis 	L3	CO3	**05**
5.	**a.**	Calculate the polar moment of inertia of a hollow circular section and radius of gyration. All dimensions are in mm. 	L4	CO4	**05**
		OR			
	b.	Find the polar moment of inertia of a solid circular section of diameter 600mm. Also find the radius of gyration.	L3	CO4	**05**
6.	**a.**	The position vector of a particle which moves in x-y plane is given by $\bar{r} = (4t^3 - 5t^2)\hat{\imath} + (0.3t^4)\hat{\jmath}$. Calculate velocity and acceleration at t = 10 sec.	L3	CO5	**05**
		OR			
	b.	A stone is thrown vertically upward and returns to ground in 6 sec. Find how high does the stone go and its initial velocity of projection.	L3	CO5	**05**
7.	**a.**	A block of mass 30 kg is placed on a plane inclined at 30° with the horizontal. A horizontal force of 250 N acts on the block tending to move the block down the plane. Find its velocity 4 sec after starting from rest. Take $\mu = 0.3$	L3	CO6	**05**
		OR			
	b.	A 500N block kept on the top of a 15° sloping surface is pushed down the plane with an initial velocity of 20 m/s. If $\mu = 0.4$, determine the distance travelled by the block and time it will take as it comes to rest. 	L3	CO6	**05**

		PART-C			**6 X 10=60**
8.	**a.**	Five forces 100N, 200N, 300N, 400N and 500N are acting at an angle of 40°,110°,210°,280° and 340° in anticlockwise direction from x-axis, at a point all acting away from the point. Calculate the resultant force in magnitude, direction and position.	L3	CO1	**10**
		OR			
	b.	Forces of 2kN, 4kN, 6kN and 8kN respectively acts on a regular pentagon as shown in the figure. Calculate the resultant in magnitude, direction and position 	L3	CO1	**10**
9.	**a.**	One end of a uniform ladder weighs 30N rest against a smooth wall and the other end on the rough horizontal floor. Coefficient of friction is 0.24. Solve for the inclination of ladder to the horizontal when it is on the point of slipping. 	L3	CO2	**10**
		OR			
	b.	Solve for the horizontal force P as shown in figure to start 400N wedge moving to the right. The angle of friction is 20° at all the contact surfaces. 	L3	CO2	**10**

10.	a.	Calculate the centroid for the shaded region. 	L3	CO3	10
		OR			
	b.	A solid cone having base diameter 15 mm and height 150 mm is placed on top of cylinder of diameter 150 mm and height 200 mm, such that axes are co-linear. Locate centre of gravity with respect to bottom.	L3	CO3	10
11.	a.	Find the polar moment of inertia with respect to reference X and Y axis for the shaded region shown in the figure. 	L3	CO4	10
		OR			
	b.	Calculate the polar moment of inertia and radius of gyration of hallow rectangular section. External dimension (300 X 200) mm and internal dimension (250 X 150) mm.	L3	CO4	10
12.	a.	A rocket is travelling straight up at 80 m/s when it runs out of fuel at height of 7000 m above the ground. Determine the maximum height reached by the rocket and the time it takes before it falls back down to the ground. Neglect the effect of air resistance.	L3	CO5	10
		OR			
	b.	An aeroplane travels on a curved path. At P it has speed of 360 km/h which is increasing at a rate of 0.5 $/s^2$. Determine at P i. The magnitude of total acceleration. ii. Angle made by the acceleration vector with positive x axis. 	L3	CO5	10

13.	**a.**	The 20 kg crate is subjected to a force having a constant magnitude F = 100 N. When s = 15 m, the crate is moving to the right with a speed of 8 m/s. Determine its speed when s = 25 m. The coefficient of kinetic friction between the crate and the ground is $\mu = 0.25$	L3	CO6	**10**
		OR			
	b.	The velocities of two identical steel blocks of mass 0.6 kg before impact are $V_A = 4\,m/s$ and $V_B = 2\,m/s$ leftward. After impact the velocity of block B is observed to be 2.5 m/s to the right, determine the coefficient of restitution between the blocks.	L3	CO6	**10**